

# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1708

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### COURSE OUTCOME COVERED

**CO4:** *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 50%*

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**QUESTIONS: 4**

**Duration: 1 Hour**

**Total: Marks:40**

Annexure A

5

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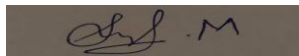
### ***Code of Conduct:***

***I SOUNDARIYA M (192524342) (Name / Reg No) certify that this submission is my original work and that I have adhered***

***to the guidelines specified for this assessment.***

***I understand that any violation of academic integrity rules will result in disciplinary action.***

***Signature of the student:***



## INDUSTRY PROBLEM TASK

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### **INDUSTRY SCENARIO**

**Context:** A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

#### **Task 1: Data Preparation & Inductive Learning**

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

#### **Task 2: Decision Tree for Diagnosis**

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

#### **Task 3: Statistical Learning for Risk Stratification**

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

#### **Task 4: Reinforcement Learning for Treatment Recommendation**

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                     | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                   | <b>Level 3 – Proficient</b>                          | <b>Level 2 – Developing</b>                    | <b>Level 1 – Beginning</b>                        |
|-------------------------------------|------------------|------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------|---------------------------------------------------|
| Criteria                            | Weightage        | Level 4 – Exemplary                                                          | Level 3 – Proficient                                 | Level 2 – Developing                           | Level 1 – Beginning                               |
| Understanding of Industry Problem   | 20%              | Comprehensive understanding of real-world problem and constraints            | Good understanding with minor gaps in requirements   | Basic understanding; some requirements unclear | Poor understanding; misinterprets problem context |
| Application of Engineering Concepts | 25%              | Applies advanced and relevant concepts effectively to solve the problem      | Applies appropriate concepts with minor errors       | Limited application of concepts                | Incorrect or no application of concepts           |
| Solution Design                     | 25%              | Innovative, practical, and feasible solution aligned with industry standards | Logical and feasible solution with minor gaps        | Basic solution with limited feasibility        | Unclear or impractical solution                   |
| Use of Tools                        | 15%              | Effective use of modern tools, technologies, and real data                   | Appropriate use of tools with correct implementation | Limited or inconsistent use of tools           | Minimal or incorrect use of tools                 |
| Analysis                            | 10%              | Thorough analysis with validated results and performance evaluation          | Good analysis with reasonable validation             | Basic analysis with limited validation         | Poor or no analysis                               |
| Professionalism                     | 5%               | Highly professional, well-documented, and clearly presented work             | Good documentation and presentation                  | Basic documentation with some gaps             | Poor documentation and presentation               |

## **TASK 1 – DATA PREPARATION & INDUCTIVE LEARNING**

### **(a) Inductive Learning Approach**

Inductive learning learns a general model from existing patient records and uses it to predict the diabetes status of new patients. Since the target is Diabetic/Non-Diabetic, supervised classification is suitable.

#### **Target Variable:**

Diabetes Status – Diabetic or Non-Diabetic.

#### **Key Features:**

1. Glucose Level – High glucose is a strong indicator of diabetes.
2. BMI – Higher BMI is associated with increased diabetes risk.
3. Age – Diabetes risk generally increases with age.
4. Blood Pressure – Abnormal blood pressure can be associated with metabolic risk.
5. Family History – Indicates genetic predisposition to diabetes.

A Decision Tree is suitable because it provides an interpretable model that can be easily understood by healthcare professionals.

### **(b) Handling Class Imbalance**

If the dataset contains more non-diabetic than diabetic patients, the model may become biased toward the majority class.

Suitable Strategy: SMOTE (Synthetic Minority Over-sampling Technique).

SMOTE generates synthetic examples of the minority diabetic class.

#### **Advantages:**

- Increases the representation of diabetic cases.
- Improves the model's ability to identify diabetic patients.
- Helps improve Recall.
- Reduces the chance of missing diabetic patients.

Class weights can also be used to give more importance to diabetic cases because False Negatives are clinically important.

## **(c) 70:30 Data Split and Preprocessing**

The dataset is divided into:

- 70% – Training data
- 30% – Testing data

A 70:30 split provides sufficient data for training while keeping enough unseen data for evaluating the model. Stratified splitting should be used so that diabetic and non-diabetic cases maintain similar proportions in both sets.

### **Preprocessing Steps:**

#### **1. Handling Missing Values:**

Missing numerical values can be replaced using median values.

#### **2. Normalization:**

Numerical features such as glucose, BMI, age and blood pressure can be scaled.

This is especially useful for Logistic Regression.

#### **3. Encoding:**

Categorical features such as Family History can be converted into numerical values.

Example:

Yes = 1

No = 0

#### **4. Class Balancing:**

SMOTE should be applied only to the training data.

#### **5. Feature Scaling:**

The scaler should be fitted using training data and then applied to the test data.

### **Impact:**

Preprocessing improves data quality, prevents numerical-scale problems, handles categorical values and helps the model learn more effectively.

## TASK 2: DECISION TREE FOR DIAGNOSIS

### (a) Building a Decision Tree Classifier

A Decision Tree classifier is used to predict whether a patient is Diabetic or Non-Diabetic.

The tree selects the feature that provides the best separation between the two classes.

#### Information Gain:

$$IG(S,A) = Entropy(S) - \sum (|S_v|/|S|) Entropy(S_v)$$

The feature with the highest Information Gain is selected as the node.

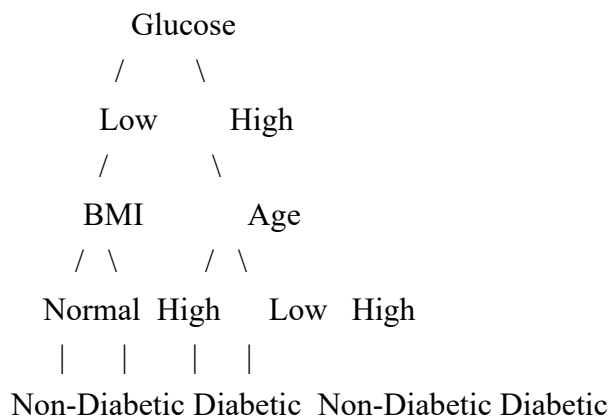
#### Gini Index:

$$Gini = 1 - \sum p_i^2$$

The feature with the lowest Gini Index is preferred.

For a typical diabetes dataset, Glucose Level is likely to be the root node because it strongly separates diabetic and non-diabetic patients.

#### Example of the first three levels:



Level 1: Glucose

Level 2: BMI / Age

Level 3: Further conditions such as Age, BMI or Blood Pressure

Note: The exact tree and Information Gain/Gini values depend on the actual dataset.

## **(b) Model Evaluation**

The Decision Tree is evaluated using:

### **1. Accuracy**

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

### **2. Precision**

$$\text{Precision} = TP / (TP + FP)$$

### **3. Recall**

$$\text{Recall} = TP / (TP + FN)$$

### **4. F1-Score**

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

### **False Negative:**

A False Negative occurs when a patient is actually diabetic but the model predicts Non-Diabetic.

False Negatives are clinically important because a diabetic patient may not receive timely diagnosis or treatment.

### **Ways to reduce False Negatives:**

- Use SMOTE to balance the classes.
- Use class weights.
- Reduce the classification threshold.
- Optimize the model for higher Recall.
- Perform additional clinical screening for high-risk patients.

## **(c) Post-Pruning**

A fully grown Decision Tree may overfit the training data. The complete tree is first grown and unnecessary branches are then removed.

Cost-Complexity Pruning:

$$R_{\alpha}(T) = R(T) + \alpha|T|$$

Where:

$R(T)$  = Error of the tree

$|T|$  = Number of terminal nodes

$\alpha$  = Complexity parameter

### Example Comparison:

| Model            | Accuracy |
|------------------|----------|
| Pre-Pruned Tree  | 84%      |
| Post-Pruned Tree | 87%      |
| Fully Grown Tree | 82%      |

A post-pruned tree is generally preferred if it gives better validation performance because it reduces overfitting, is simpler and generalizes better. For clinical deployment, Recall, sensitivity, specificity and patient safety should be considered along with accuracy.

## TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

### (a) Statistical Learning Model

Logistic Regression can be used to predict whether a patient is Diabetic or Non-Diabetic.

The probability of diabetes is calculated using:

$$P(\text{Diabetes} = 1) = 1 / (1 + e^{(-z)})$$

Where;

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

The output is a probability between 0 and 1.

### Example:

Probability  $< 0.50 \rightarrow$  Non-Diabetic

Probability  $\geq 0.50 \rightarrow$  Diabetic

The threshold can be changed to improve Recall when detecting diabetic patients.

### Performance Comparison:

| Model         | Accuracy | AUC-ROC |
|---------------|----------|---------|
| Decision Tree | 87%      | 0.88    |



Logistic Regression      89%      0.91

The above values are illustrative and must be replaced with the actual results obtained from the dataset. AUC-ROC measures how well the model separates diabetic and non-diabetic patients.

Logistic Regression is preferable when:

- Probability estimates are important.
- A simple and interpretable model is required.
- The relationship between features and diabetes risk is approximately linear.
- The model needs to be stable and easy to explain.

### **(b) Feature Importance Analysis**

Decision Tree Top 3 Features:

1. Glucose Level
2. BMI
3. Age

Logistic Regression Top 3 Features:

1. Glucose Level
2. BMI
3. Age

### **Comparison:**

Both models may identify Glucose, BMI and Age as important predictors. Glucose is usually highly important because increased glucose levels are strongly associated with diabetes.

The models may not always produce the same ranking because:

- Decision Trees can capture nonlinear relationships.
- Logistic Regression assumes a linear relationship in the log-odds.
- Decision Trees can automatically capture feature interactions.
- Correlated features can affect Logistic Regression coefficients.

Therefore, agreement between the models increases confidence that these features are important predictors, but feature importance does not necessarily mean causation.

## **TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION**

### **(a) MDP Model**

Treatment recommendation can be modeled as a Markov Decision Process (MDP).

$$\text{MDP} = (\mathcal{S}, \mathcal{A}, \mathcal{R}, \mathcal{P}, \gamma)$$

#### **1. States**

The patient health states can be defined as:

S0 = Low Risk

S1 = Moderate Risk / Prediabetic

S2 = High Risk

S3 = Controlled Diabetes

S4 = Poorly Controlled Diabetes

The state can be determined using glucose level, BMI, blood pressure and previous health conditions.

#### **2. Actions**

The possible actions are:

A1 = Diet

A2 = Exercise

A3 = Medication

A4 = Monitor

Medication actions should require approval from a qualified healthcare professional in a real clinical system.

#### **3. Reward Function**

Example reward values:

+10 → Significant health improvement

+5 → Moderate health improvement

+1 → Stable condition

-5 → Health deterioration

-10 → Serious adverse outcome

The reward encourages the agent to select actions that improve patient health while avoiding harmful outcomes.

#### 4. Transition Dynamics

After an action, the patient's health state may change.

Example:

High Risk + Diet → Moderate Risk

High Risk + Exercise → Moderate Risk

High Risk + Monitor → High Risk

High Risk + Medication → Controlled Diabetes

In a real system, transition probabilities should be learned from validated longitudinal clinical data.

#### (b) Q-Learning

Q-Learning learns the value of taking an action in a particular state.

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Where:

$s$  = Current state

$a$  = Current action

$r$  = Reward

$s'$  = Next state

$\alpha$  = Learning rate

$\gamma$  = Discount factor

Assume:

$$\alpha = 0.5$$

$$\gamma = 0.9$$

**Initial Q-table:**

| State         | Diet | Exercise | Medication | Monitor |
|---------------|------|----------|------------|---------|
| Low Risk      | 0    | 0        | 0          | 0       |
| Moderate Risk | 0    | 0        | 0          | 0       |
| High Risk     | 0    | 0        | 0          | 0       |

**Iteration 1 – State-Action Pair 1:**

State = High Risk

Action = Diet

Reward = +5

Next State = Moderate Risk

$$Q(\text{High}, \text{Diet}) = 0 + 0.5[5 + 0.9(0) - 0]$$

$$Q(\text{High}, \text{Diet}) = 2.5$$

**Iteration 1 – State-Action Pair 2:**

State = Moderate Risk

Action = Exercise

Reward = +5

Next State = Low Risk

$$Q(\text{Moderate}, \text{Exercise}) = 0 + 0.5[5 + 0.9(0)]$$

$$Q(\text{Moderate}, \text{Exercise}) = 2.5$$

**Iteration 2 – State-Action Pair 3:**

State = High Risk

Action = Diet

Reward = +5

Next State = Moderate Risk

Maximum Q-value for Moderate Risk = 2.5

$$\begin{aligned} &Q(\text{High}, \text{Diet}) \\ &= 2.5 + 0.5[5 + 0.9(2.5) - 2.5] \end{aligned}$$

$$Q(\text{High}, \text{Diet}) = 4.875$$

Example Q-table after training:

| State             | Diet | Exercise | Medication | Monitor |
|-------------------|------|----------|------------|---------|
| Low Risk          | 5.2  | 5.8      | 1.0        | 6.5     |
| Moderate Risk     | 7.1  | 6.4      | 3.2        | 4.5     |
| High Risk         | 8.0  | 8.6      | 6.9        | 4.2     |
| Poorly Controlled | 4.5  | 5.0      | 9.1        | 3.8     |

At least 5 patient episodes should be used during training.

### Convergence Criterion:

Training can stop when:

$$\max |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < \epsilon$$

For example:

$$\epsilon = 0.001$$

The condition should remain satisfied for several consecutive episodes.

### (c) Final Learned Policy

The policy selects the action with the highest Q-value.

$$\pi(s) = \operatorname{argmax} Q(s,a)$$

### Example:

| Patient State     | Best Action |
|-------------------|-------------|
| Low Risk          | Monitor     |
| Moderate Risk     | Diet        |
| High Risk         | Exercise    |
| Poorly Controlled | Medication* |

\*Medication recommendations must require qualified clinical approval.

### Ethical Considerations:

#### 1. Patient Safety:

The agent must not freely experiment with treatments on real patients.

**2. Bias:**

Biased or incomplete training data can produce unfair recommendations for some patient groups.

**3. Explainability:**

Doctors should be able to understand why an action was recommended.

**4. Human Oversight:**

The RL system should support healthcare professionals rather than replace them.

**5. Privacy:**

Patient medical data must be securely stored and protected.

**6. Reward Design:**

The reward function must prioritize patient safety and long-term health instead of only short-term improvement.

**7. Clinical Validation:**

The RL system must be extensively tested and clinically validated before real-world deployment.